

AMRITA VISHWA VIDYAPEETHAM, AMRITAPURI

B. Tech CSE (ARTIFICIAL INTELLIGENCE)

22AIE499-LITERATURE REVIEW SUBMISSION

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4. Problem Definition:

The objective of Phase 1 was to study and evaluate neural approaches for Sanskrit Sandhi splitting, a core preprocessing task in Sanskrit Natural Language Processing. Phase 1 involved a linguistic study of Sandhi, a review of rule based and neural approaches, and the implementation of character level neural models including sequence to sequence and two stage architectures. The models were evaluated using standard metrics and detailed error analysis was conducted. The study highlighted key limitations of purely neural methods, particularly in low resource settings and in handling Sanskrit's rule governed grammatical structure. Based on feedback provided by the review panel, Phase 2 focuses on the implementation and systematic evaluation of neural models and modern large language models for Sanskrit Sandhi splitting. This phase emphasizes systematic benchmarking of multilingual and low resource models and cross language comparison to highlight Sanskrit's unique linguistic characteristics. It also includes detailed qualitative and quantitative error analysis and the exploration of Panini inspired rule based or hybrid approaches to address the limitations observed in purely neural and large language model based systems.

5. Literature Survey Summary Table

No	Authors & Proceedings	Year	Journal & Impact Factor	Methodology / Approach	Limitations & Future works
1	Idicula, S. M. (2024). Review on Sanskrit Sandhi Splitting using Deep Learning Techniques. Journal of Information Technology, 6(2), 136-152.	2024	JITDW (IF $\approx$ 1.2)	Survey of neural and DL-based sandhi methods	Lacks direct benchmarking on LLMs; identifies need for standardized datasets.
2	Ali, I., Presti, L. L., Spano, I., & La Cascia, M. (2025). ABBIE: Attention-Based BI-Encoders for Predicting Where to Split Compound Sanskrit Words. ICAART, 2, 334-344.	2025	ICAART (Core B)	Bi-encoder neural model with attention for split prediction	Struggles with multi-word recursive compounds; code yet to be validated on Vedic data.
3	Nehrdich, S., Hellwig, O., & Keutzer, K. (2024). One model is all you need: ByT5-Sanskrit, a unified model for Sanskrit NLP tasks. arXiv preprint arXiv:2409.13920.	2024	arXiv	Byte-level Transformer: Pretraining on Sangraha dataset for joint Segmentation and POS tagging.	High latency for long sentences; sensitivity to non-standard orthography.
4	Sandhan, J., Agarwal, A., Behera, L., Sandhan, T., & Goyal, P. (2023). Sanskritshala: A neural sanskrit nlp toolkit with web-based interface for pedagogical and annotation purposes. arXiv preprint arXiv:2302.09527.	2023	ACL Demo	Neural Toolkit: Web-based interface integrating BiLSTM models for segmentation and parsing.	Error analysis is limited to model accuracy; lacks LLM comparative metrics.
5	Maheshwari, A., Singh, N., Krishna, A., & Ramakrishnan, G. (2022). A benchmark and dataset for post-OCR text correction in Sanskrit. arXiv preprint arXiv:2211.07980.	2022	EMNLP	Seq2Seq: Byte-level tokenization with phonetic encoding (ByT5+SLP1) for correction.	Primarily focused on OCR; Sandhi splitting is a secondary task here.

6	Pradeep, A., & Mamidi, R. (2025). Sandarśana: A Survey on Sanskrit Computational Linguistics and Digital Infrastructure for Sanskrit. ACM Computing Surveys, 57(10), 1-38.	2025	ACM CSUR (IF: 16.0)	Evaluates digital infrastructure and neural trends over the last decade	Qualitative only; does not propose a specific new sandhi architecture.
7	Kulkarni, A., Goyal, P., Kumar, S., & Shrawar, K. (2024). Error patterns in Sanskrit morphological processing. arXiv. doi.org	2024	arXiv	Systematic taxonomy of errors in neural segmenters	No implementation of a fix; identifies gaps for hybrid models.
8	Hu, J., et al. (2023). How Well Do LLMs Handle Morphologically Rich Languages?	2023	ACL	Cross-lingual evaluation of GPT models on low-resource scripts.	Sanskrit is treated as a generic low-resource language; ignores Sandhi rules.
9	Chang, L., et al. (2024). Evaluating Multilingual LLMs on Low-Resource Classical Languages.	2024	arXiv	Analysis of Llama and GPT-4 on Sanskrit literary tasks.	Limited linguistic depth regarding phonetic transformations.
10	Kulkarni, A. (2024). Integrating Paninian Grammar with Neural Models.	2024	WSC	Using Paninian Sutras as logic constraints for neural splitters	Scalability issues with complex rules; tested on small datasets.
11	Tandra, S., et al. (2023). Building Linguistic Resources for Sanskrit NLP.	2023	Workshop	Focus on gold-standard data for sandhi and morphology.	Resource-centric; lacks deep neural modeling experiments.
12	Reddy, R., et al. (2023). Sequence Modeling Challenges in MRLs.	2023	arXiv	Benchmarking Transformer vs. RNN for highly inflected languages	General focus; not specific to the Sanskrit Sandhi phenomenon.
13	Sharma & Nirkhe (2024). Rule-Guided Neural Models for Sanskrit Word Segmentation.	2024	arXiv	Penalizing neural predictions that violate Paninian constraints.	Lacks comparison with modern LLM-based prompting.
14	Feng, Y., et al. (2023). Beyond Accuracy: Error Analysis Frameworks for NLP.	2023	EMNLP	Systematic qualitative analysis protocols for	Not language-specific; provides the "method" rather than the "solution."

				deep learning models.	
15	Maheshwari et al. (2024). LEVOS: Leveraging Vocabulary Overlap for Sanskrit Lexicon Generation.	2024	arXiv	Using lexicon overlap to improve neural segmentation in low-resource data.	Performance drops significantly on archaic Vedic structures.

6. Identified Research Gaps:

Tokenization Barriers: Current LLMs (GPT-4, Llama) use subword tokenization that splits Sanskrit characters in ways that are phonetically nonsensical according to Paninian grammar.

Absence of Hybrid Benchmarking: While "Rule-Guided" models exist (Sharma, 2024), there is no systematic study that benchmarks **modern LLMs** (Phase 2 goal) against these hybrid approaches.

Qualitative Failure Analysis: Most papers provide "Accuracy" or "F1-scores," but few provide a linguistic deep-dive into *why* models fail at specific Sandhi boundaries (e.g., *Visarga* vs. *Vyanjana*).

7. Novelty / Contribution of the Proposed Work

Systematic LLM Benchmarking: Evaluating high-resource multilingual LLMs specifically for Sandhi splitting.

Neuro-Symbolic Module: Implementing a hybrid layer where the LLM proposes splits, but a **Panini-inspired validator** (based on traditional *Sūtras*) acts as a logical filter to ensure linguistic validity.

Cross-Language Context: Comparing Sanskrit to other morphologically rich languages to highlight the specific challenges posed by phonetic fusion.

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